

Subject Description Form

Subject Code	ABCT3102
Subject Title	Commercialization of Biotechnology Products
Credit Value	2
Level	3
Pre-requisite	ABCT2101 Biochemistry or ABCT2102 Molecular Biology or ABCT2103 Cell Biology
Co-requisite	Nil
Exclusion	Nil
Objectives	To understand the process of development and commercialization of biotechnology products. To understand the patent process and the process of regulatory approval.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: - Understand the process of drug development from bench work to preclinical to clinical trial - Understand what is intellectual property and its importance in biotechnology - Understand the process of regulatory approval for biotechnology products - Write a business plan for a start-up biotechnology company
Subject Synopsis/ Indicative Syllabus	Process of developing a drug: from bench to a product including Product Life Cycle Management and tools (14 hours) • <i>In vitro</i> activity • <i>In vivo</i> activity • Pharmacokinetics (ADME) and toxicity • Pre-clinical trial • Clinical trial phase I, II and III • Cost estimation of each stage Innovation (4hrs)

	Importance of patents in biotechnology (4 hours) • Different kinds of patents: Utility, Design, Plant, or Provisional Patent. • Writing Descriptions, Claims, or Abstracts for a patent application • Brief introduction to different regulatory bodies around the world. Business plan (4 hours) • Purpose of a business plan. • Different components of a business plan • Writing your own business to raise funding for a start-up biotech company
Teaching/Learning Methodology	Lecture, Tutorial, Mini-Project: preparation and presentation of a business plan.

Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks		% (weighting)	Intended subject learning outcomes to be assessed (Please tick as appropriate)				
				a	b	c	d	e
	1. Attendance		10	√	√			
	2. Tutorial discussion		20	√	√			
	3. Mid-term written report		25	√	√	√	√	√
	4. Final project of business plan		45	√	√	√	√	√
	Total		100 %					
	(a) knowledge (b) teamwork (c) creativity (d) writing quality (e) logical flow							
	Students are required to attend at least 75% of scheduled sessions for the subject. Students fail to fulfill the attendance requirement will lose the 10% attendance score.							
	Student Study Effort Expected	Class contact:						
▪ Lecture					20 Hrs.			
▪ Tutorial					6 Hrs.			
Other student study effort:								

	▪ Assignment	10 Hrs.
	▪ Self study	28 Hrs.
	Total student study effort	64 Hrs
Reading List and References	1. Business Development for the Biotechnology and Pharmaceutical Industry, By Martin Austin. Routledge (Publisher). 2. Drugs, From Discovery to Approval, By Rick Ng. Wiley-Blackwell, 2nd Ed. 2009- Online library	